Enhancing Vietnamese Hate Speech Detection with Multi-agent Debate

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Abstract. Hate speech detection is a key element of maintaining a healthy discourse environment for social media platforms, particularly for low-resource languages like Vietnamese. In our paper, we propose a new approach drawing inspiration from multi-agent debate to improve the accuracy of discrimination between toxic and non-toxic comments. The system consists of two independent large language model agents that initially produce independent decisions, then undergo a formal debate arguing their sides, and a neutral judge agent finally produces the decision. We try the system out on the dataset of the ViCTSD and demonstrate that our approach attains 89.80% accuracy and 79.66% macro F1-score, outperforming the standard single-model baselines. Our experiments demonstrate that the incorporation of the debate scheme helps the agents better put the content in context and reduces the misclassification arising from ambiguous phrasing or latent aggressivity. Our paper demonstrates a potential direction towards leveraging cooperative AI agents to enhance the reliability and interpretability of systems for the moderation of toxic content.

Keywords: Hate speech detection · Multi-agent debate · Large language models · Vietnamese NLP · Multi-agent systems

1 Introduction

The sudden proliferation of social media over the last few years has been accompanied by the troubling emergence of hate speech, which seriously impairs the ability to have a constructive and secure online discussion. For the Vietnamese language case, identifying the existence of such hate speech becomes even difficult due to the usage of informal linguistic expressions such as slang, abbreviation, code-switching, and implicit sarcasm, which complicate automatable content moderation systems as problematic. Classical approaches employed to date have predominantly employed pretrained language models

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belonging to the BERT family, such as PhoBERT, BERTweet, and XLM-R. These models have shown promising results, their effectiveness often depends heavily on task-specific techniques such as text normalization, data augmentation (e.g., Easy Data Augmentation – EDA), or specialized loss functions (e.g., Focal Loss) to address label imbalance [1]. While useful, such techniques are resource inefficient, laborious, and lack generalizability over the realm/languages. To alleviate these limitations, the use of large language models (LLMs) has become a highly attractive alternative. By virtue of their categorization capability, LLMs are strong natural language reasoning and inferring context models, which prefer them the best structural deliberation candidates.

In our paper, we propose a new Vietnamese hate speech detection framework, which adopts a Multi-Agent Debate (MAD) framework, in which a set of LLM-based agents, all independent, categorize a comment, exchange arguments, and, through a judge agent, collectively make a judgment. Based on the inspiration of the paper of the realm of debate theory as well as multi-agent reasoning, a design aims at facilitating a better classification accuracy as well as improved interpretability for borderline/ambiguous instances. The primary goal of the current paper is to explore the potential of enabling LLM agents to "debate" such that the system better identifies hate speech when faced with subtle hate speech, contextual nuances, or controversial remarks.

The main contributions of this paper are as follows:

- We introduce a novel classification architecture based on MAD using LLMs, specifically tailored for Vietnamese hate speech detection.
- We experimented with the system over the UIT-ViCTSD [2] dataset, a widely used standard for Vietnamese toxic comment categorization, to ensure the system's robustness under realistic conditions.
- We demonstrate that the debate mechanism reduces misclassification by leveraging opposing perspectives and formalized arguing among agents, such that the system can be scaled and constructed to provide explainable AI moderation systems for low-resource tongues.

The remainder of this paper is structured as follows: Sect. 2 reviews related work, Sect. 3 describes the proposed MAD model, Sect. 4 presents experiments and results, and Sect. 5 concludes with future directions.

2 Related Work

Hate speech detection constitutes a significant challenge in the field of natural language processing (NLP), particularly for low-resource languages such as Vietnamese. The linguistic characteristics of Vietnamese, including the use of slang, code-switching, and implicit sarcasm, further complicate automatic classification efforts. Initial studies mostly utilized traditional machine learning models such as Support Vector Machines, Random Forest, and Naive Bayes, with hand-engineered features such as TF-IDF and n-grams. Despite this, traditional approaches have proved ineffective in modeling rich semantics as well as context-dependent subtleties characteristic of social media speech.

Transformer-based models have significantly improved Vietnamese hate speech classification tasks specially, those in the BERT family. Special-purpose Vietnamese pre-trained language models such as PhoBERT [3] and BERT4News [4], along with multilingual variants such as XLM-R [5], have extensively been proved successful on standard datasets such as UIT-ViCTSD [2]. In addition, those using CNNs in a way of blending them with those pre-trained models represented by those such as PhoBERT-CNN [6] as well as those in ensemble schemes, have realized macro F1-scores usually in between 65% and 74%. Despite these breakthroughs, there are issues in dealing with sophisticated linguistic structures as well as in borderline hate speech content.

Another long-standing problem in Vietnamese hate speech corpora is significant class skewness, with only a small fraction of toxic comments being present in the datasets. In response, data Augmentation strategies such as Vietnamese-adapted EDA using WordNet and fastText [1], as well as Focal Loss [7], have proven successful in reducing this skewness. These strategies, however, tend to be tailored using language-dependent preprocessing libraries such as VnCoreNLP [8], thus being restricted in transfer across other domains or languages. LLMs such as GPT-3.5, GPT-4, and GPT-4o [9], have exhibited good classification and reasoning performance, with powerful baselines in hate speech identification without fine-grained feature engineering or with no need for language-dependent libraries. Most current studies, however, still utilize single-model outputs, with no internal validity checks or uncertainty handling—a vital requirement in dealing with subjectivity and vagueness in hate speech classification. Multi-agent systems employing debate-based reasoning have been identified as a promising direction. The MAD framework presented at EMNLP 2024 introduced structured argumentation among multiple agents under a judge's supervision, fostering divergent reasoning processes and improving performance in complex reasoning tasks [10]. Recent studies have explored the integration of adversarial debate strategies and voting mechanisms within multi-agent systems to reduce hallucinations and enhance decision reliability in large language model applications [11].

This paper introduces a Vietnamese hate speech classification system based on a MAD framework, where multiple LLMs independently analyze input texts, engage in structured debate, and reach final classification decisions through adjudication. By combining diverse perspectives and internal validation, this approach enhances classification accuracy, interpretability, and reliability, offering a solid alternative to traditional single-agent methods.

3 Proposed Model

3.1 Multi-agent Debate Architecture

Figure 1 provides a comprehensive illustration of the operational mechanism behind the proposed debate-based classification model, structured around a MAD framework consisting of two independent language agents and one neutral adjudicator. The classification process begins with an initialization phase, where both agents independently receive the same input comment for classification. Each agent operates based on a customized prompt that includes the comment to be classified, contextual information such

as the topic and title of the related article, and step-by-step reasoning instructions following a chain-of-thought approach. Based on these prompts, both agents independently produce an initial label—either "toxic" or "non-toxic" along with a detailed explanation reflecting their individual interpretive perspectives.

The system then proceeds to the first debate round, which takes place regardless of whether the two agents initially agree or disagree. During this round, both agents directly respond to each other's arguments, either reinforcing their own position or challenging the opponent's reasoning. Their interaction is subsequently evaluated by the neutral Judge Agent, according to these key criteria: internal logical consistency, alignment with contextual evidence and overall persuasiveness. The Judge Agent provides qualitative feedback, identifies the more reliable agent (TrustedAgent), and proposes a provisional label (ChosenLabel). If both agents produce the same label and their reasoning is deemed sufficiently robust, the system may proceed directly to the final decision stage.

If disagreement persists after the first round, the debate continues into a second and final round. Both agents have the opportunity to revise their stance and present new arguments, drawing on feedback from the Judge Agent, the opponent's reasoning, and the original comment context. This process ensures that any changes in position are the result of rational inference rather than arbitrary adjustments. At the conclusion of the second round, if consensus is reached, the majority label is confirmed, provided that the Judge Agent validates the soundness of the supporting arguments. If no consensus is achieved, the Judge Agent issues the final classification decision based on an overall assessment of argument quality, prioritizing the label accompanied by the most convincing explanation, typically from the TrustedAgent.

The entire decision-making process culminates in the final classification and explanation stage. The Judge Agent determines the definitive label and selects one representative argument as the official explanation. This multi-layered adjudication process is designed not only to enhance classification accuracy but also to ensure transparency, traceability, and alignment with content moderation requirements in socially sensitive environments such as Vietnamese-language social media platforms.

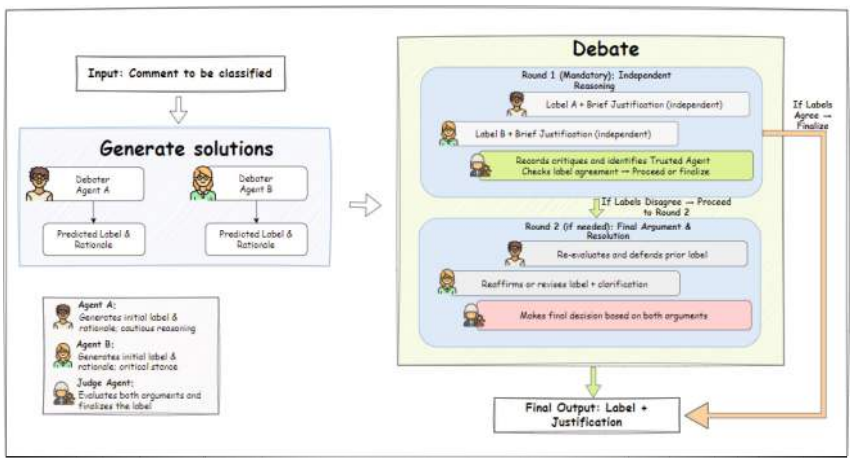

Fig. 1. Overview of the MAD framework for Vietnamese hate speech detection

3.2 Agent Style Design

As a help to the multi-agent discussion framework, we created a role-based system of cueing that assigns a particular perspective and reasoning ability to a particular agent. This approach seeks to optimize the reasoning and analysis abilities of the model while ensuring consistency and interpretability as the model goes through the process of categorization. Every language agent gets a varying prompt with a given cognitive bias to recognize hate speech.

Notably, the two primary classifier agents are established as follows:

- Agent A adopts a cautious and neutral evaluation style, only assigning the label "toxic" when there is clear evidence of profanity, insult, or direct attack.
- Agent B demonstrates heightened sensitivity to indirect offense, group-based discrimination, and implicit harmful language, placing greater emphasis on social context.

A detailed description of the agent styles is illustrated in Table 1.

Table 1. Agent style design for MAD in Vietnamese hate speech classification.

Agent	Description
Agent A	You are a social media comment analyst with a neutral and rigorous mindset. You only assign the 'toxic' label when there is sufficient evidence of profanity, explicit insults, or direct attacks. You prioritize precision and objectivity, avoiding mislabeling comments that are mildly sarcastic, express personal opinions, or present moderate criticism. If a comment is not clearly offensive or harmful, you should retain the 'non-toxic' label"

(*continued*)

Table 1. (*continued*)

Agent	Description
Agent B	You are a social media comment analyst with high sensitivity to indirect insults, group discrimination, and biased language. You emphasize social context and are cautious of comments that may contain covertly harmful or prejudiced implications. Even in the absence of explicit profanity, if a comment expresses extreme negativity, overgeneralization, or heavy sarcasm targeting vulnerable groups, you may consider labeling it as 'toxic'

3.3 Inter-Round Debate Procedure and Consensus Mechanism

After the two Debater Agents complete their initial arguments, the system initiates a supervised multi-round debate process under the coordination of a neutral adjudicator to ensure a transparent and rigorous classification workflow. In the first round (t = 0), each agent produces a classification label from the set {toxic, non-toxic}, along with a detailed justification reflecting its analytical perspective. The Judge Agent evaluates these arguments based on three criteria: internal logical consistency, completeness and contextual relevance (including connections to the original comment and reference examples), and overall persuasiveness.

Regardless of whether the agents initially agree or disagree on their labels, the first debate round is always conducted. During this round, each agent is required to respond to the other's argument either reinforcing, challenging, or adjusting its reasoning. Based on the exchange, the Judge Agent provides qualitative feedback (Critique), identifies the more trustworthy agent (TrustedAgent), and proposes a provisional label (ChosenLabel).

If both agents agree on the classification label in the first round, the system proceeds to the final evaluation phase. At this stage, the Judge Agent confirms the agreed label if both accompanying rationales meet the quality standards; otherwise, the label may be revised if the consensus is not well-justified. If disagreement remains after the first round, the system advances to a second debate round (t = 1). In this final round, each agent may either maintain or revise its position, based on three sources of information: the Judge Agent's feedback, the opponent's prior argument and rebuttal, and the original comment context.

To model how agents update their stances, a weighted adjustment formula is applied to the numeric encoding of labels. Specifically, each agent's new label score $S_i^{(1)}$ is calculated as:

$$S_i^{(1)} = \alpha.S_i^{(0)} + (1 - \alpha).S_j^{(0)} \tag{1}$$

where $\alpha \in [0,1]$ reflects the agent's level of self-confidence, and the label encoding is defined as: non-toxic = 0, toxic = 1.

We conducted a sensitivity analysis for $\varepsilon \in [0.3–0.4]$, showing performance varied within $\pm$ 0.8%. An ablation study showed that removing Eq. (1) reduced Macro F1 from 79.66 to 77.4, validating its contribution.

Consensus is measured in a binary fashion: it is considered achieved if both agents output the same label (value $= 1$) and not achieved if they disagree (value $= 0.5$). The system is said to reach consensus when the agreement score is greater than or equal to $1 - \varepsilon$, with $\varepsilon = 0.34$, effectively requiring both agents to produce identical labels.

If consensus is achieved after the second round, the Judge Agent validates the label, provided both justifications meet the reasoning standards. If one argument is clearly weaker, the system favors the label associated with the stronger argument. If no consensus is reached after two rounds, the final decision rests with the Judge Agent, who issues the classification based on a holistic assessment of the persuasiveness of both arguments. In situations where no compelling rationale clearly outweighs the other, the system defaults to the label supported by the TrustedAgent deemed more reliable during the debate. We experimented with extending the debate to three rounds. While this setting improved Macro F1 by only 0.2%, it increased inference time by approximately 40%. Therefore, we limit the debate to two rounds, which offers a practical trade-off between accuracy and computational efficiency.

In each round, the Judge Agent's feedback consistently comprises three components: a critique aligned with standardized evaluation criteria, the most reasonable provisional label at that point, and identification of the agent with the more convincing rationale. To ensure neutrality, the Judge Agent was explicitly instructed in the prompt to (i) evaluate only logical consistency and persuasiveness, (ii) not rely on majority voting, and (iii) avoid emotional reasoning. For example, in Round 1 the Judge prompt was: 'You are a neutral Judge Agent…'. In Round 2: 'You are the final Judge Agent… synthesize arguments without favoring any agent.' Role-swapping experiments confirmed $< 1\%$ variation, supporting neutrality.

4 Experiments

4.1 Dataset

Table 2. Representative comments from the UIT-ViCTSD dataset used in Vietnamese hate speech classification.

Comment	Title	Topic	Label
Bale hậu vệ cánh trái,Son hậu vệ cánh phải là chuẩn bài Mou nổ rồi *(Bale as left back, Son as right back—that's the perfect setup. Mou will explode with joy)*	Bale trở về Tottenham *(Bale returns to Tottenham)*	Sports	non-toxic
Bản chất là côn đồ thôi, chỉ chờ cơ hội là bộc phát. *(It's essentially just thuggery, merely waiting for an opportunity to erupt)*	Ba cô giáo bị dừng công tác vì không bảo vệ trẻ *(Three teachers were suspended for failing to protect the children.)*	Education	toxic

We conducted our experiments on the UIT-ViCTSD [2] dataset, a benchmark corpus for the task of Vietnamese hate speech detection. The dataset consists of a total of 10,000 user-generated comments collected from VnExpress one of the largest online news platforms in Vietnam and each comment was manually labeled as either toxic or non-toxic. Some representative examples are shown in Table 2. The original dataset is notably imbalanced, with toxic comments accounting for only about 10.8% of the total samples.

We used the official test set the dataset, which contains 1,000 samples, including 890 non-toxic comments and 110 toxic comments, strictly adhering to the original label distribution to ensure consistency with the dataset's inherent class proportions and to provide a realistic evaluation scenario reflecting the natural imbalance observed in real world data. Furthermore, we did not apply any data augmentation, oversampling, or class-weight adjustments during sampling. All comments were preserved in their original form, without any preprocessing steps such as lowercasing, tokenization, special character removal, or stop word filtering.

4.2 Experimental Setup

The experiments in this paper were conducted via API interaction with the Gemma 3 27B Instruct model provided by Google as a cloud-based service, utilizing the model through an API to access the strong natural language reasoning capabilities of LLMs without requiring parameter fine-tuning or local deployment; all experiments were run on a personal computer with the following hardware specifications: OS: Windows 10 Pro, CPU: Intel® Xeon® Platinum 8272CL @ 2.60 GHz, RAM: 64 GB DDR4 ECC, GPU: NVIDIA RTX 4060 Ti (16 GB VRAM). Although we used the Gemini API, the underlying model Gemma-3-27B-IT is open-source and available on Hugging Face. It can be replicated locally with sufficient hardware (e.g., NVIDIA A100 80 GB, H100 80 GB, or multi-GPU V100 32 GB).

4.3 Compared Methods

To evaluate the effectiveness of the proposed MAD system, we conducted a comparative analysis against a range of representative baseline methods, including BERT (cased), PhoBERT, BERT4News, XLM-R, DistilBERT [1] and Mistral 7B Instruct [12]. All baseline models were evaluated using the macro-F1 score, consistent with the evaluation criteria of our system, to ensure objectivity and fairness in comparison.

4.4 Evaluation Metrics

We employ the following metrics to evaluate model performance:

- F1-macro is the average of F1-scores computed independently for each class, balancing precision and recall.

$$F1 - macro = \frac{1}{N} \sum_{i=1}^{N} F1_i \tag{2}$$

where N is the number of classes, and $F1_i$ is the F1-score of class i, calculated as follows:

$$F1_i = \frac{2.Precision_i.Recall_i}{Precision_i + Recall_i} \tag{3}$$

- Precision is the proportion of predicted positives that are actually positive:

$$Precision_i = \frac{TP_i}{TP_i + FP_i} \tag{4}$$

- Recall is the proportion of actual positives that are correctly identified:

$$Recall_i = \frac{TP_i}{TP_i + FN_i} \tag{5}$$

Where TP is True Positive, FP is False Positive, TN is True Negative, FN is False Negative.

4.5 Results and Discussion

Table 3. Performance comparison between the proposed model and baseline methods

Model	Method	Toxic Precision (%)	Toxic F1 (%)	Non-toxic Precision (%)	Non-toxic F1 (%)	Macro F1 (%)	Accuracy (%)
BERT	BERT (cased)	–	–	–	–	68.69	-
	PhoBERT	–	–	–	–	74.63	-
	BERT4News	–	–	–	–	74.58	-
	DistilBERT	–	–	–	–	69.67	-
XLM-R	XLM-RoBERTa	–	–	–	–	73.57	-
LLMs	mistral-7b-instruct-v0.3-bnb-4bit (fine-tune)	65.43	64.78	93.80	86.02	75.40	91.50
	Gemma-3-27b-instruct (zero-shot)	36.88	41.43	93.25	91.60	66.51	85.30
Proposed Model	Gemma-3-27b-instruct (MAD)	52.17	65.31	98.28	94.02	79.66	89.80

Table 3 presents the performance comparison between the proposed MAD model and several baseline methods. The proposed system achieved a Macro F1 score of 79.66% and an accuracy of 89.8%, outperforming both conventional BERT-based models such as PhoBERT, BERT4News, and XLM-R, as well as LLMs in terms of F1-macro. The model maintains a balanced performance across both toxic and non-toxic classes, avoiding the tendency of single-pass classifiers to favor one class disproportionately. This observation is reflected in the confusion matrix of the proposed method on the VICTSD dataset (see Fig. 2), where 96 out of 110 toxic samples and 802 out of 890 non-toxic samples were correctly identified. We additionally conducted bootstrap resampling (1,000

iterations) to validate statistical significance. The improvement of MAD over PhoBERT and BERT4News is significant at $p < 0.05$.

The confusion matrix further reveals a system preference for non-toxic labels, particularly in scenarios where Agent A—the more conservative debater—plays a dominant role in the decision-making process. This preference helps minimize false positives when predicting hate speech but also results in missed detections of subtle toxic comments, especially those expressed through indirect language, sarcasm, or cultural idioms. Specifically, 14 toxic samples were misclassified as non-toxic, most of which lacked explicit attack markers and instead involved emojis, abbreviations, or context-specific slang. When compared to baseline models, BERT-family architectures such as PhoBERT and BERT4News achieved Macro F1 scores ranging from approximately 73% to 74.6%, but their performance depends heavily on language-specific preprocessing steps like Vietnamese word segmentation using VnCoreNLP or synthetic data augmentation. The extra steps involved in retraining and manual data adjustments tend to make deployment across different datasets or domains more complicated and harder to scale. In contrast, the MAD system relies purely on prompt-based inference, eliminating the need for retraining and making it both more flexible and cost-efficient to deploy. That said, reviewing misclassified examples reveals ongoing challenges, especially with content that carries cultural nuances, slang, or symbols like emojis—common features in Vietnamese social media that even advanced classifiers struggle with. Closing these gaps may require adding external resources such as slang dictionaries, knowledge graphs, or expanding the debate process to allow multi-turn reasoning for deeper context handling. A key strength of the MAD model lies in its explainability. While standard transformer models often behave like black boxes, this system records the back-and-forth debate between agents and the final decision by the judge, allowing clear traceability of how classifications are made.

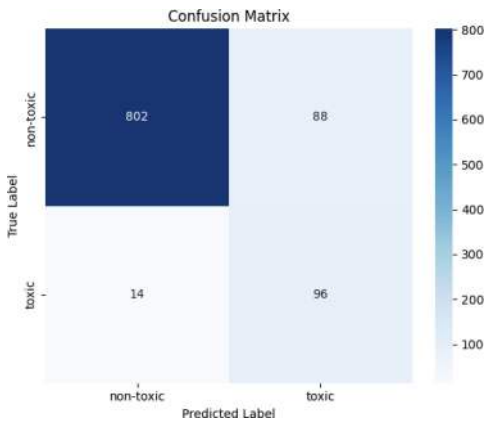

Fig. 2. Confusion matrix of the proposed method on the VICTSD dataset

4.6 Efficiency and Resource Usage

On average, initializing one agent required ~15 s, and a full two-round debate (two agents + judge) required ~3 min per comment. Token usage per comment: initialization 1,600 tokens, debate round 1: 2,663, debate round 2: 2,643. In total, ~6,906 prompt tokens were sent and ~8,271 tokens generated, totaling ~15,177 tokens. Processing 1,000 comments required ~50 h and ~15M tokens. While slower than PhoBERT (~0.4 s/comment), MAD offers interpretability and improved precision–recall balance.

4.7 Error Analysis

Table 4. Misclassified examples

Number	Comment	Ground truth	Predict
1	Nhìn cụ thế kia ko đụng cụ cháu cũng tự gục *(Looking at that elder, even without a touch, the guy would collapse on his own)*	non-toxic	toxic
2	Sao BMW mà xấu đau đớn thế này?!. *(How can a BMW be this painfully ugly?!)*	toxic	non-toxic

Several borderline cases during classification highlight the system's shortcomings in recognizing implicit aggression and subtle language nuances. Table 4 presents specific misclassified examples that reflect these limitations, particularly with understated cues and context-dependent humor. Take, for example, the comment: "Sao BMW mà xấu đau đớn thế này?!" ("How can a BMW be this painfully ugly?!"). Agent A classified it as non-toxic, reasoning that the statement simply expresses a personal opinion in an exaggerated emotional tone about a product's appearance, without targeting any individual or group. Agent B considered it an indirect form of verbal attack, as the expression "xấu đau đớn" ("painfully ugly") uses strong language that could offend or hurt BMW enthusiasts—a clearly defined user community. Nevertheless, the system's final decision favored the non-toxic label, overlooking the potentially offensive tone embedded in what appears to be harsh criticism. This case illustrates a blind spot in handling figurative or implicitly biased statements.

On the other hand, the comment "Nhìn cụ thế kia ko đụng cụ cháu cũng tự gục" ("Looking at that elder, even without a touch, the guy would collapse on his own") carries a mildly sarcastic and astonished tone, without using profane language or directly attacking a specific person or group. Within the context of a legal news story where a 67-year-old woman single-handedly defeats a male neighbor, this comment more likely conveys amazement or amusement rather than intent to insult. However, both agents labeled it as toxic, and the Judge Agent failed to counter that interpretation, resulting in a misclassification. All three agents seemed to overinterpret the tone of the statement and neglected the humorous expression, which is common in Vietnamese online discourse. This error further highlights the current debate mechanism's inadequacy in distinguishing lighthearted jokes from truly harmful speech. These two cases underscore the need

to refine evaluation criteria and prompt design to enhance the system's ability to handle ambiguous comments and language patterns typical of social media interactions.

5 Conclusion

This paper presents a novel Vietnamese hate speech detection approach using a MAD framework powered by LLMs. Unlike conventional fine-tuning methods, our system employs independent agents that classify, debate, and synthesize reasoning to deliver explainable final decisions. Operating fully in inference mode without retraining or task-specific preprocessing, the system outperforms baseline models including BERT-based and instruction-tuned LLMs on the UIT-ViCTSD dataset, achieving the highest F1-macro score. It offers both strong performance and enhanced explainability through structured argumentation, making it well-suited for socially sensitive content moderation. Future work includes developing multi-turn, context-aware debates, integrating external knowledge, applying contrastive learning for minority classes, and adding lightweight post-hoc explanation modules for greater transparency in real-world deployment. This paper has several limitations, including reliance on a single dataset and computational overhead. We partially mitigated these through stratified cross-validation, prompt-level neutrality, and efficiency analyses. Future work should extend MAD to multi-dataset settings and explore optimizations for real-time moderation.

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